

Analyzing Marketing Campaigns with Pandas

Project Report | Marketing Analytics in Python (DataCamp)

Project Objective

This project applies pandas and exploratory data analysis to a subscription business's marketing dataset to answer three core business questions: how a campaign performed, which channel is referring the most subscribers, and why a particular channel is underperforming. The goal is to move beyond simple reporting toward actionable, data-driven recommendations.

Dataset Overview

The dataset contains 10,037 user records spanning marketing touchpoints from House Ads, Email, Facebook, Instagram, and Push notifications. Each record captures the channel, the date served, the message variant shown, whether the user converted, and (for converted users) their subscription and retention status.

Key Findings

Channel	Conversion Rate	Subscribers Acquired
Email	34.16%	193
Instagram	14.16%	265
Facebook	12.74%	237
Push	8.36%	83
House Ads	6.30%	298

1. Overall campaign performance: The campaign converted 10.74% of users served across all channels, with conversion fluctuating notably over the campaign period (see Figure 2).

2. Top referring channel: Email has by far the highest conversion rate (34.16%) despite a smaller served audience, making it the most efficient channel. House Ads drove the highest raw subscriber count (298) but converts at the lowest rate (6.30%), meaning its volume is driven by reach rather than effectiveness.

3. Underperforming channel — House Ads: Breaking conversion down by day of week shows House Ads stays consistently low (5–7%) across every day, unlike Email, which spikes on weekends (over 80%). By age group, House Ads converts best among 24–30 year olds (8.52%) and worst among 30–36 year olds (4.43%), suggesting its messaging and targeting are not resonating with the core older demographic it reaches.

4. Retention: Of users who converted, 66.76% were retained, indicating that while acquisition varies sharply by channel, once a user subscribes, retention is reasonably strong regardless of source.

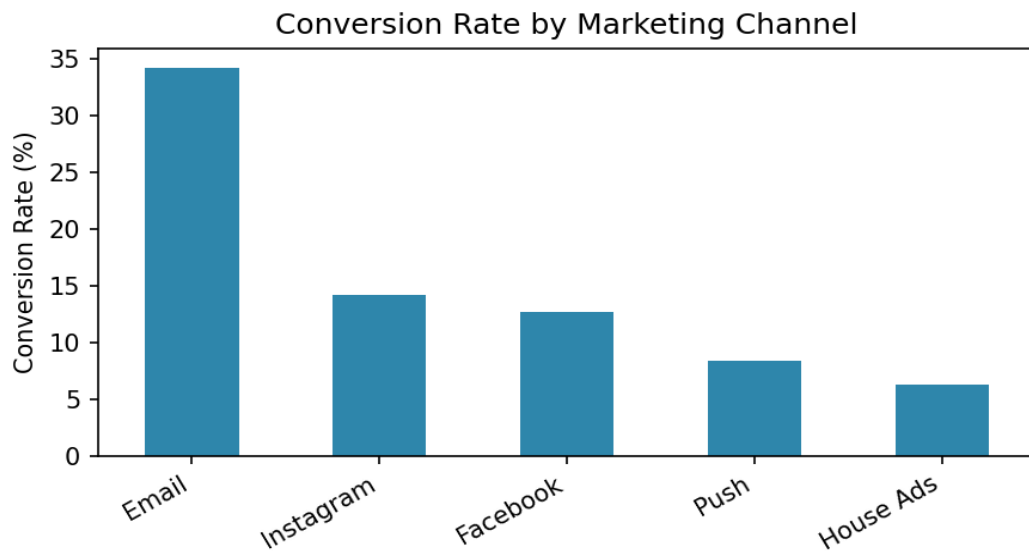


Figure 1: Conversion rate by marketing channel.

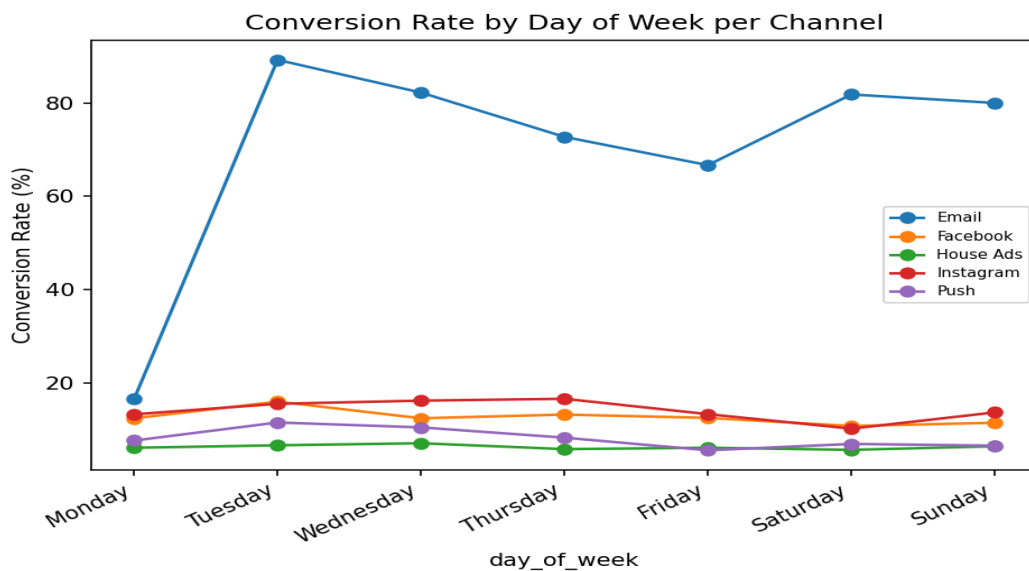


Figure 2: Conversion rate by day of week, per channel — Email's weekend spikes vs. House Ads' flat, low performance.

Recommendations

Reallocate budget toward Email for conversion-focused campaigns, given its conversion rate is over 5x that of House Ads.

Revise House Ads targeting and creative for the 30–55+ age brackets, where conversion is weakest, or shift House Ads spend toward the 24–30 age group where it performs comparatively better.

Time Email sends around weekends to capture the observed conversion spike, and investigate whether House Ads' flat day-of-week pattern reflects a lack of day-specific targeting or creative rotation.

Treat House Ads as a reach/awareness channel rather than a conversion driver, since its strength is volume (298 subscribers) rather than efficiency.